

Slide 1: Centralized Database vs Distributed Ledger

This slide is comparing two different ways of managing data.

The headings are:

1. **Input**
2. **Storage**
3. **Processing**
4. **Output**
5. **Control**

And at the bottom there is a legend showing:

- **single entity control**
- **multiple entity control**
- **single node**
- **multiple nodes**

So the whole purpose of this slide is to show:

In a centralized database, one main authority controls everything.

In a distributed ledger, multiple parties and multiple nodes are involved.

1) What happens in a centralized database?

In a centralized database:

Input

Data may come from multiple sources, but it usually enters into a system controlled by one central authority.

For example:

- students submit fee data
- staff submit attendance data
- a customer submits payment info

Even if many people give data, the system collecting it is still centrally managed.

Storage

The actual stored database is usually in one central place or under one main owner's control.

That does **not always literally mean one physical machine**, but conceptually it means:

- one authority owns the database
- one party decides what is stored
- one party has final control

Processing

Any updates, queries, verification, correction, deletion, etc. are handled centrally.

For example:

- the bank updates balances
- the university updates CGPAs
- the company updates payroll

So the logic is centralized.

Output

The results are also given from that same central system.

For example:

- bank statement
- transcript
- transaction history
- employee report

Control

This is the most important part.

In a centralized database, **one entity controls the whole process**.

That means:

- one organization decides what is valid
- one organization can modify the record
- one organization can stop access

- one organization becomes the trusted authority
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2) What is the weakness of centralized control?

This slide is indirectly pointing toward the weaknesses:

Single point of control

One authority controls the truth.

Single point of failure

If that authority fails, gets hacked, shuts down, or acts unfairly, the whole system is affected.

Need for trust

Everyone must trust that one party to:

- store correctly
- process fairly
- not manipulate data
- stay available

That is how traditional systems usually work.

And that is fine in many cases. Sir is not saying centralized databases are useless. He is saying blockchain is different because it removes that total central dependence.

3) Then what is a distributed ledger?

A distributed ledger changes this model.

Instead of:

- one main controller
- one main trusted record
- one main processing point

we have:

- multiple participating entities

- multiple nodes
- shared data
- shared maintenance
- shared agreement

So the key difference is not only “data is copied in many places.”

The deeper difference is:

control and trust are distributed rather than concentrated.

That is the real point of this slide.

Slide 2: DLT — An Accurate Description

This slide starts explaining distributed ledger technology in a much more correct way.

It says:

- Consider four entities/nodes distributed in the network
- Transaction data is generated by these nodes and shared
- Each node builds a **journal**
- Its own version of the database
- For this example: 4 journals
- Journals could be different; why?
- The blockchain is the unified view of individual journals

This slide is extremely important.

1) Four nodes in the network

The picture shows Node A, Node B, Node C, and Node D connected with each other.

This means:

there are multiple participants in the system, not one central server.

Each node:

- receives information
- validates information
- processes it

- relays it onward

The arrows in the picture are showing that data is being shared among them.

So blockchain is not “one machine maintaining one record.”

It is a network.

2) Transaction data is generated by the nodes

This means the participants themselves create the activity.

For example:

- Node A creates a transaction
- Node C creates another
- Node D receives and relays it
- Node B validates it too

So data is not only being entered into one office or one database manager's software.

It is being generated and shared by distributed participants.

That is why distributed systems are harder than centralized ones.

3) Each node builds a journal

This is the most important concept here.

A **journal** means:

each node keeps **its own local record** of what it believes the shared data currently is.

So:

- Node A has Journal A
- Node B has Journal B
- Node C has Journal C
- Node D has Journal D

Each journal is that node's own view of the system.

You should think of a journal like a notebook.

Imagine 4 students in a class all trying to keep attendance records themselves.

There is no teacher collecting one master register.

Each student writes attendance in their own notebook.

Those notebooks are the journals.

4) Why can journals be different?

This is a super important question from the slide.

If all journals are supposed to represent the same shared reality, then why can they differ?

Because this is a distributed network, and in distributed systems, information does **not** reach everyone at the exact same instant.

Journals can differ because:

a) Network delay

Node A may receive a transaction first, but Node C may receive it later.

b) Temporary offline state

One node may be disconnected for a while.

c) Validation timing

One node may still be checking whether some data is valid.

d) Different local order of receipt

Two nodes may hear about two transactions in different order.

So when the slide says journals could be different, it means:
temporary mismatch is natural in distributed systems.

That is a very important exam idea.

A lot of students wrongly assume distributed ledger means every node always has perfectly identical data every second.

No.

The goal is not perfect instant sameness.
The goal is convergence and synchronization.

5) “The blockchain is the unified view of individual journals”

This is the core line of the slide.

It means the blockchain is not just:
“one big database sitting somewhere.”

Instead, blockchain is the **common agreed version** that emerges from all the individual journals.

So each node has its own local record, but through communication and agreement, those local records move toward one shared accepted version.

That shared accepted version is what we call the blockchain or distributed ledger.

This is a very deep point.

Simple way to remember it:

- Journal = one node’s local notebook
 - Blockchain = the shared agreed story formed from all notebooks
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Slide 3: DLT — An Accurate Description (Part 2)

This slide continues the same idea and says:

- Goal: Different journals are in sync **most of the time**
- Offer a **shared view**
- Sync effort must be without a central party
- What is the ledger?
 - Not the database
 - Mechanism of syncing different journals

This is one of the most conceptual slides in Week 1.

1) Goal: journals are in sync most of the time

Notice the wording carefully:

it does **not** say “all journals are perfectly identical every second.”

It says:

most of the time

Why?

Because in real distributed systems:

- messages take time
- nodes can lag
- new data may still be spreading
- not all nodes update instantly

So perfect immediate sameness is unrealistic.

What blockchain wants is:

- enough synchronization
 - enough agreement
 - enough consistency
- so that the network practically behaves as one shared system

That is why “most of the time” is a realistic systems statement.

2) Offer a shared view

Even though each node has its own journal, users of the system need one answer to questions like:

- Who owns this asset?
- Did this payment happen?
- Which block is the latest accepted one?

That single answer is the **shared view**.

Without a shared view, the system becomes useless because different nodes would keep telling different stories.

So the whole purpose of distributed ledger tech is to make many local records behave like one trusted common record.

3) Sync effort must be without a central party

This line is the heart of blockchain.

If there *were* a central party, syncing would be easy:
everyone would just copy whatever the central authority says.

But blockchain wants synchronization **without** saying:

“the bank decides,”

or

“the company server decides,”

or

“the government server decides.”

That means the nodes must coordinate and agree among themselves.

That is why blockchain is both powerful and difficult.

4) What is the ledger?

The slide gives a very subtle answer:

The ledger is not the database

This is really important.

At first, students think:

ledger = stored table of transactions.

But sir is trying to make you think more accurately.

In distributed ledger systems, the ledger is not just “a database file.”

The ledger is the mechanism of syncing different journals

This means the ledger is really the **shared agreed state** that comes out of the synchronization process.

So the word “ledger” here means something more than storage.

It means:

- shared state
- synchronized view
- agreed record
- outcome of consensus

Best way to understand this

Each node has its own journal.

The ledger is what exists when these journals are kept aligned through synchronization.

So the ledger is not one notebook.

It is the agreed result of keeping many notebooks in sync.

That is exactly what the picture on the right is trying to show with the journals around the nodes and the circular arrows.

The circular arrows mean:

keep syncing, keep aligning, keep updating.

Slide 4: Syncing — Building consensus

This slide says:

- Set of journals
 - Individual views of shared data
- Convergence to a single shared view
 - That everyone agrees with — the distributed ledger
- A ledger isn't quite a tangible thing
 - Just emerges from the efforts of synchronizing individual journals
- Distributed consensus (Syncing): How does that work?
 - Five-step process, but before that:
 - Building distributed consensus is HARD!

This is basically the conclusion of the previous two slides.

1) Set of journals = individual views of shared data

This means every node starts with its own local perspective.

Each node sees:

- what it has received
- what it has validated
- what it has recorded so far

So each journal is one participant's current understanding of the network's shared data.

The important phrase is:

individual views of shared data

That means they are all trying to track the same reality, but from separate local positions.

2) Convergence to a single shared view

This is the most important phrase on the slide.

Convergence means coming together.

So even if journals start slightly different, the system wants them to move toward one common accepted version.

That common accepted version is:

the distributed ledger

So the distributed ledger is not just “many copies.”

It is:

many local views converging into one shared agreed view.

That is a much stronger and more accurate definition.

3) “That everyone agrees with”

This is the actual meaning of consensus.

Consensus means:
the nodes agree on what the correct shared history is.

For example:

- everyone agrees that Muneeb paid Zain
- everyone agrees that Block 57 is valid
- everyone agrees that Alice's transaction is real
- everyone agrees that the latest chain looks a certain way

That agreement is what gives blockchain credibility.

Without agreement, there is no single truth, only conflicting local records.

4) “A ledger isn’t quite a tangible thing”

This line confuses students, but it is actually simple.

It means:
the ledger is not one physical object you can point at and say “this exact file is the ledger.”

In centralized systems, you can often point to one database and say:
that is the master record.

But in a distributed ledger system, the ledger is more like:

- the shared agreed state
- the synchronized result
- the common reality formed by many nodes

So the ledger is not fully tangible as one single database object.
It **emerges** from coordination.

That is why the next line says:

“Just emerges from the efforts of synchronizing individual journals”

That means when all the nodes keep sharing, validating, processing, and aligning their journals,
the common ledger appears as the result.

So the ledger is an **emergent property** of synchronization.

That sounds fancy, but the simple meaning is:
the ledger comes into existence because many nodes keep their journals aligned.

5) Building distributed consensus is HARD

This is the final and very important message.

Why is it hard?

Because:

- there is no central boss
- nodes may receive data at different times
- some nodes may fail
- some nodes may be malicious
- journals may temporarily differ
- the network still has to agree on one valid shared version

That is exactly why blockchain needs carefully designed consensus mechanisms.

So this slide is basically preparing you for later lectures:

“Okay, now that you understand what the goal is, we will next study how the network actually achieves that goal.”

Full story of these 4 slides together

Let me connect all 4 slides as one story so you remember them properly.

In a **centralized database**, even if data comes from many sources, one authority controls storage, processing, output, and truth. That is simple, but it creates dependency on one central party.

In a **distributed ledger**, there is no single controller. Instead, many nodes participate. These nodes generate and share transaction data. Each node keeps its own **journal**, meaning its own local version of the shared database. Since the system is distributed, these journals may not always match perfectly. Different nodes may receive information at different times, validate at different speeds, or temporarily go offline.

The goal of DLT is to keep these journals **in sync most of the time** so that the network offers one **shared view**. This syncing must happen **without a central authority**. That is why the ledger is not simply one database file. The ledger is really the **mechanism and result of synchronizing different journals**.

When many individual journals gradually move toward one agreed version, that process is called **convergence**, and the agreed version is the **distributed ledger**. This shared agreed state “emerges” from the efforts of synchronization. Achieving this agreement among many distributed participants is called **distributed consensus**, and it is hard because no central authority exists and the participants may have delays, differences, or faults.

What sir can ask from these slides

These 4 slides are very short in text, but sir can make many conceptual questions from them.

He can ask:

1. Differentiate centralized database and distributed ledger

Answer should focus on:

- central vs distributed control
- single trusted authority vs shared agreement
- single point of failure vs reduced central dependence

2. What is meant by a journal in DLT?

Answer:

- each node’s own local version of the shared data
- one node’s notebook/record
- may temporarily differ from other journals

3. Why can journals differ?

Answer:

- network delays
- offline nodes
- different validation timing
- different order of receiving information

4. What is meant by saying “the blockchain is the unified view of individual journals”?

Answer:

- blockchain is the shared agreed version
- formed from many local records
- not just one central database

5. Why does the slide say “the ledger is not the database”?

Answer:

- because ledger is not just stored data
- it is the synchronized agreed state
- it emerges from syncing different journals

6. What is convergence in DLT?

Answer:

- many local journals moving toward one common accepted view
- agreement formation
- basis of distributed ledger

7. Why is distributed consensus hard?

Answer:

- no central authority
 - delays/failures/malicious nodes
 - still need one agreed truth
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Ratta-friendly definitions from these slides

Centralized database

A centralized database is a system where one main authority controls the storage, processing, output, and truth of the data.

Distributed ledger

A distributed ledger is a shared record maintained across multiple nodes where agreement is achieved without a central controlling authority.

Journal

A journal is a node's own local version of the shared database or transaction record.

Shared view

A shared view is the common accepted version of the data that the participating nodes agree on.

Ledger

In DLT, the ledger is not merely a database; it is the shared agreed state that results from syncing different journals.

Convergence

Convergence is the process by which many individual journals move toward one single shared agreed view.

Distributed consensus

Distributed consensus is the process through which multiple nodes in a distributed network agree on one common version of the shared data.

One-line memory trick

Remember these 4 slides like this:

Centralized system = one master database.

DLT = many local journals.

Blockchain = unified agreed view of those journals.

Consensus = the process of keeping those journals aligned.